

SIFFI SINGH

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SUMMARY

Applied Scientist with 7+ years building production-scale NLP, Generative AI, multilingual AI, and conversational analytics systems across AWS and Alexa. Specialized in LLM applications, semantic retrieval, speech and language understanding, and research-to-production AI systems serving thousands of enterprise customers globally. Published at ACL, NAACL, EMNLP, EACL, COLING, and CIKM with experience leading end-to-end AI launches, mentoring scientists, and driving cross-functional product development.

EXPERIENCE

Applied Scientist II

May 2022 – Present

Amazon Web Services

Pasadena, US

- Led development and launch of Automated Theme Detection for 4,000+ Amazon Connect customers including Barclays, Air Canada, and Expedia, enabling businesses to identify emerging customer issues across large-scale call and chat conversations using Generative AI, semantic clustering, multilingual NLP, LLM-as-a-judge evaluation, and LLM-based data augmentation for testing, achieving 86% cluster coherence and reducing manual conversation analysis time by 80%.
- Led deployment of Semantic Intent Retrieval and AI-Powered Summarization, replacing keyword search with semantic retrieval using vector embeddings and intent understanding, achieving 70% precision and reducing customer search and discovery time from hours to minutes across voice and chat domains.
- Expanded multilingual AI capabilities across 14 languages by building PII Redaction using LLM-generated synthetic data, XLM-RoBERTa, and code-switching detection, achieving an average macro F1 score of 80% across languages while improving safety and privacy for multilingual customer conversations and documents.
- Built a real-time Automatic Speech Translation prototype for AWS Transcribe supporting direct speech-to-text translation for multilingual communication with English–Spanish translation achieving BLEU score of 60.

Applied Scientist

Sep 2019 – May 2022

Amazon Alexa

Cambridge, UK

- Built large-scale information extraction systems for Alexa by training relation extraction and ontology alignment models from web tables; created RedTab, a benchmark dataset later adopted for production evaluation of knowledge graph extraction models.
- Scaled Alexa’s Evi Knowledge Graph by 130% (44.5M to 103M entities) by building distributed ingestion pipelines for 11M structured facts across IMDB Movies and People while improving answer freshness through stale-fact removal.
- Developed multilingual question-answering systems combining entity linking, knowledge graphs, and language model reasoning across English, French, Spanish, Japanese, and Hindi.

Machine Learning Research Scientist

Jun 2017 – Jul 2019

Industrial Technology Research Institute / JMicron Technology

Hsinchu, Taiwan

- Commercially launched LivCore, a privacy-preserving face clustering and recognition product achieving 86% ARI accuracy using lightweight TensorFlow models optimized for constrained hardware and real-world face variations.
- Developed multimodal benchmark datasets for image, video, and LiDAR fusion for autonomous vehicle perception systems and implemented real-time classification algorithms on DSP hardware.

SELECTED PUBLICATIONS

- [1] HP Zou, **S Singh**, Y Nian, et al. “**Active Generalized Category Discovery with Diverse LLM Feedback**,” *European Chapter of the Association for Computational Linguistics (EACL)*, 2026.
- [2] I Shalymov, H Su, **S Singh**, et al. “**Controllable Conversational Theme Detection Track at DSTC 12**,” *Dialog System Technology Challenge (DSTC 12)*, 2025.
- [3] Y Zhang, **S Singh**, S Sengupta, et al. “**Can Your Model Tell a Negation from an Implicature? Unravelling Challenges with Intent Encoders**,” *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- [4] H Aboutaleb, **S Singh**, et al. “**MAGID: An Automated Pipeline for Generating Synthetic Multi-Modal Datasets**,” *North American Chapter of the ACL (NAACL)*, 2024.
- [5] L Tang, I Shalymov, **S Singh**, et al. “**TofuEval: Evaluating Hallucinations of LLMs on Topic-Focused Dialogue Summarization**,” *North American Chapter of the ACL (NAACL)*, 2024.

- [6] H Song, I Shalyminov, **S Singh**, et al. “**Enhancing Abtractiveness of Summarization Models through Calibrated Distillation,**” *Findings of Empirical Methods in NLP (EMNLP)*, 2023.
- [7] KH Huang, **S Singh**, X Ma, et al. “**SWING: Balancing Coverage and Faithfulness for Dialogue Summarization,**” *Findings of the European Chapter of the ACL (EACL)*, 2023.
- [8] **S Singh**, AF Aji, GS Tomar, C Christodoulopoulos. “**A Relation Extraction Dataset for Knowledge Extraction from Web Tables,**” *International Conference on Computational Linguistics (COLING)*, 2022.
- [9] Y Zhou, **S Singh**, C Christodoulopoulos. “**Tabular Data Concept Type Detection Using Star-Transformers,**” *ACM International Conference on Information and Knowledge Management (CIKM)*, 2021.
- [10] **S Singh**, HM Hang. “**A Continuous Learning System for Face Clustering and Recognition,**” *IEEE International Conference on Consumer Electronics (ICCE)*, 2021.

EDUCATION

National Chiao Tung University	Hsinchu, Taiwan
<i>M.S. Electrical Engineering and Computer Science, GPA: 4.0/4.0</i>	<i>2017 – 2019</i>
Dr. A.P.J. Abdul Kalam Technical University	Noida, India
<i>B.Tech Computer Science and Engineering, Ranked 1st among 10,000+ students</i>	<i>2013 – 2017</i>

TECHNICAL SKILLS

- **Languages:** Python, Java, Scala, SQL, C/C++
- **ML / LLMs:** PyTorch, TensorFlow, HuggingFace Transformers, Sentence-Transformers, RAG, Fine-tuning, Prompt Engineering, LLM Evaluation, Semantic Search, Vector Embeddings
- **Cloud / Systems:** AWS SageMaker, Lambda, EC2, S3, DynamoDB, EMR, Glue, Step Functions, Docker, PySpark
- **Domains:** Conversational AI, Multilingual NLP, Speech Recognition, Knowledge Graphs, Summarization, Information Extraction, Synthetic Data Generation

HONORS AND LEADERSHIP

- **External Collaborator, California Institute of Technology** (Prof. Sergei Gukov) – Research on reinforcement learning and search algorithms for mathematical reasoning and combinatorial optimization, 2025–2026
- **External Collaborator, University of Cambridge** (Prof. Burigede Liu) – Research on material property prediction using LLMs and tabular deep learning models, 2024–2025
- **Organizer, DSTC 12 Controllable Conversational Theme Detection Track**, 2025
- **Research Mentor, Amazon Alexa & AWS**, 2019–2026 – Guided scientists and engineers whose work led to publications at ACL, EMNLP, and EACL
- **Best Paper Award (1st Place), IEEE ICCE**, 2021
- **Invited Speaker** - “Making Alexa More Knowledgeable,” AI & Big Data Expo and REWORK AI Summit London, 2022
- **Silver Medal, ACM ICPC Asia Jakarta Regionals**, 2017
- **Google Women Techmakers Scholar**, 2018